

Question# 01

Calculate from the following data gave the growth of employment in lakhs in organized sector between 1989 and 1995

- a) Correlation Coefficient r
- b) Equation of regression of line when $x = 5$

Year	1988	1989	1990	1991	1992	1993	1994	1995
Public Sector	98	101	104	107	113	120	125	128
Private Sector	65	65	67	68	68	69	68	68

Question# 02

- a) In testing a certain kind of truck tire over rugged terrain, it is found that 25% of the trucks fail to complete the test run without a blowout. Of the next 15 trucks tested, find the probability that
 - (i). from 3 to 6 have blowouts
 - (ii). fewer than 4 have blowouts
 - (iii). more than 5 have blowouts
- b) National security requires that defense technology be able to detect incoming projectiles or missiles. To make the defense system successful, multiple radar screens are required. Suppose that three independent screens are to be operated and the probability that any one screen will detect an incoming missile is 0.8. Obviously, if no screens detect an incoming projectile, the system is unworthy and must be improved.
 - (i). What is the probability that an incoming missile will not be detected by any of the three screens?
 - (ii). What is the probability that the missile will be detected by only one screen?
 - (iii). What is the probability that it will be detected by at least two out of three screens?

Question# 03

- a) On average, 3 traffic accidents per month occur at a certain intersection. What is the probability that in any given month at this intersection,
 - (i). Exactly 5 accidents will occur
 - (ii). Fewer than 3 accidents will occur
 - (iii). At least 2 accidents will occur?

- b) If the probability that a fluorescent light has a useful life of at least 800 hours is 0.9, find the probabilities that among 20 such lights,
- (i). exactly 18 will have a useful life of at least 800 hours
 - (ii). at least 15 will have a useful life of at least 800 hours
 - (iii). at least 2 will not have a useful life of at least 800 hours.
- c) The probability that a student pilot passes the written test for a private pilot's license is 0.7. Find the probability that a given student will pass the test
- (i). on the third try
 - (ii). before the fourth try.

Question# 04

- a) The total number of hours, measured in units of 100 hours, that a family runs a vacuum cleaner over a period of one year is a continuous random variable X that has the density function,

$$f(x) = \begin{cases} x; & 0 < t < 1 \\ 2 - x; & 2 < t < 3 \\ 0; & \text{elsewhere} \end{cases}$$

Find the probability that over a period of one year, a family runs their vacuum cleaner

- (i). less than 120 hours
 - (ii). between 50 and 100 hours.
- b) The waiting time, in hours, between successive speeders spotted by a radar unit is a continuous random variable with cumulative distribution function,

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-ax}, & x \geq 0 \end{cases}$$

Find the probability of waiting less than 12 minutes between successive speeders

- (i). using the cumulative distribution function of X
- (ii). using the probability density function of X

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Question # 05

- a) The loaves of rye bread distributed to local stores by a certain bakery have an average length of 30 centimeters and a standard deviation of 2 centimeters. Assuming that the lengths are normally distributed, what percentage of the loaves are

(i). Longer than 31.7 centimeters

(ii). Between 29.3 and 33.5 centimeters in length

(iii). Shorter than 25.5 centimeters

- b) A research scientist reports that mice will live an average of 40 months when their diet are sharply restricted and then enriched with vitamins and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 6.3 months find the probability that a given mouse will live

(i). more than 32 months

(ii). less than 28 months

(iii). between 37 and 49 months

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$$f(x) = \begin{cases} x_1 & 0 < t < 1 \\ 2 - x_1 & 2 < t < 3 \\ 0, & \text{elsewhere.} \end{cases}$$

Find the probability that over a period of one year, a family runs their vacuum cleaner (a) less than 120 hours; (b) between 50 and 100 hours.

- b) The waiting time, in hours, between successive speeders spotted by a radar unit is a continuous random variable with cumulative distribution function,

$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-8x}, & x \geq 0 \end{cases}$$

Find the probability of waiting less than 12 minutes between successive speeders (a) using the cumulative distribution function of X ; (b) using the probability density function of X .

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